

Banking Transaction Risk & Customer Insights Dashboard

Project Overview

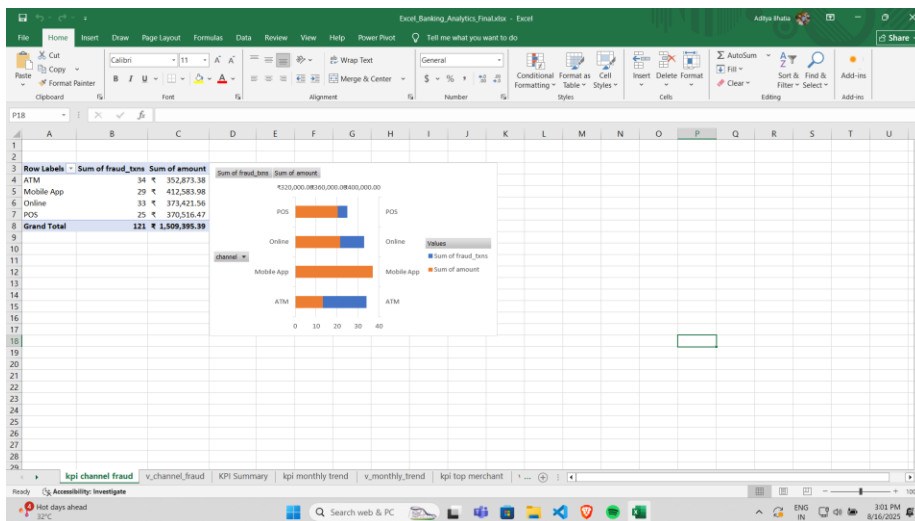
This project focuses on analyzing banking transactions to identify patterns, risks, and business opportunities. The system integrates Python and SQL for (data storage & cleaning), Excel (pivot-based summaries), and Power BI (interactive dashboard) to provide a complete end-to-end solution. The goal is to deliver insights into fraud detection, customer behavior, and regional performance that can guide decision-making in the banking sector.

Key Objectives

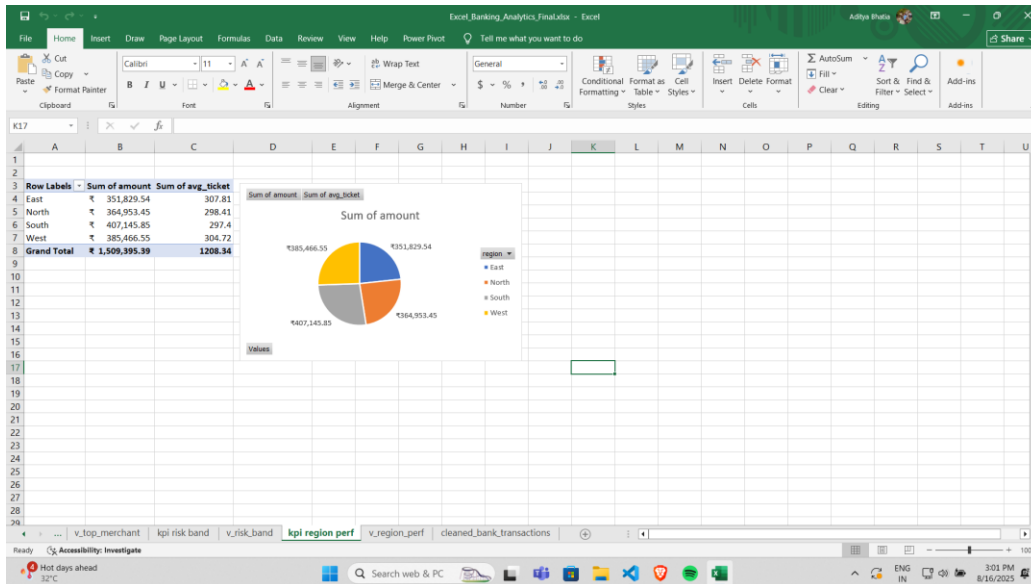
1. Detect and analyze fraudulent transactions across different channels.
2. Track monthly transaction trends and ticket sizes for business monitoring.
3. Assess regional performance to highlight strong and weak areas.
4. Segment customers into risk bands to identify high-risk categories.
5. Identify top merchants contributing most to transaction volume.

Meaningful Business Insights

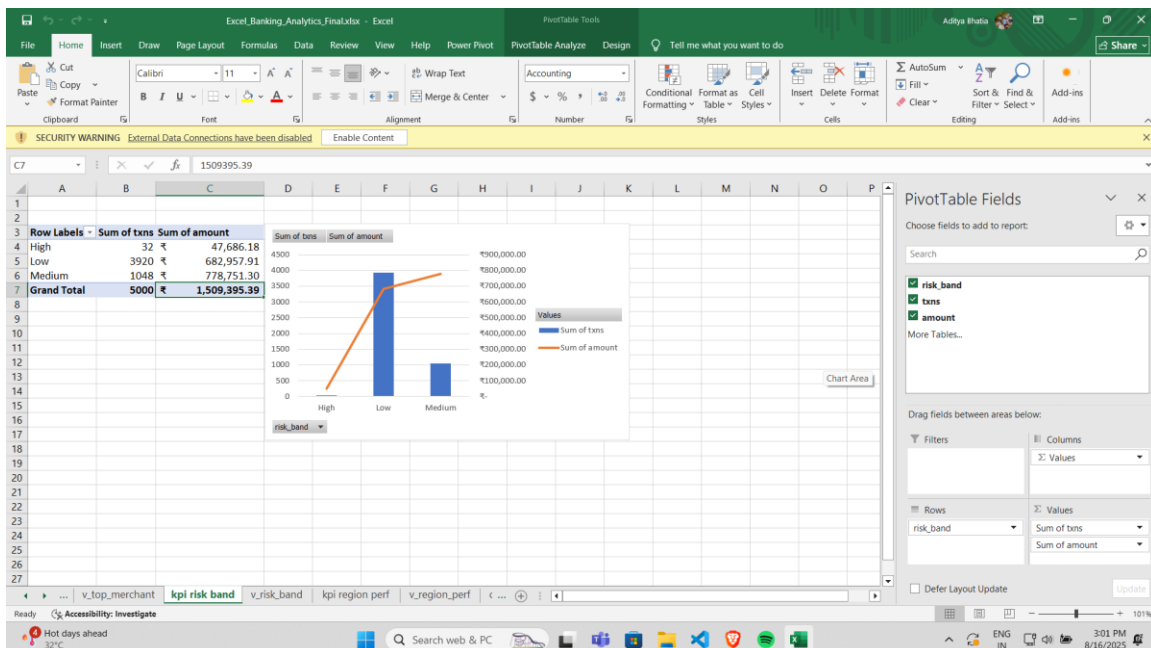
- **Fraud Analysis:** The dashboard highlights channels with the highest fraud risk, enabling banks to strengthen controls and focus monitoring efforts.



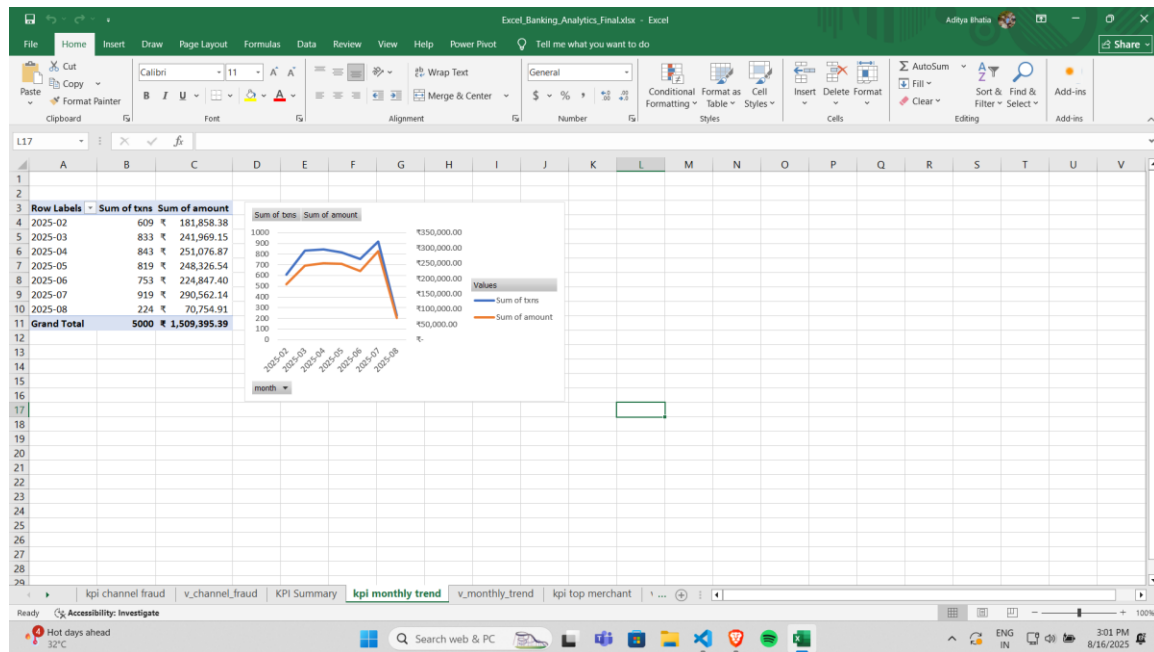
- **Regional Performance:** Regional breakdown helps identify profitable and underperforming markets, guiding targeted marketing and resource allocation.



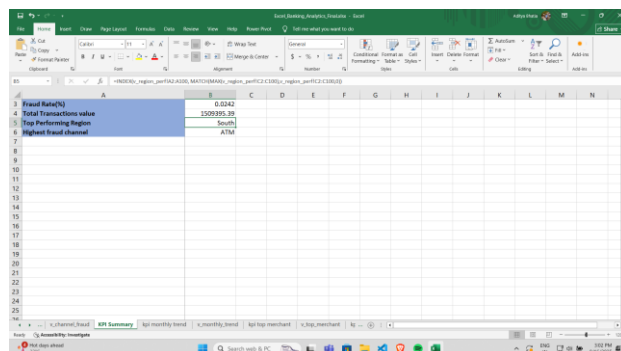
- **Risk Band Distribution:** By categorizing transactions into low, medium, and high-risk bands, the system supports better risk management and customer profiling.



- **Monthly Trends:** Time-based analysis allows monitoring of transaction growth, seasonality, and emerging risks over time.



- **Kpi Summary:**



Tools & Technologies Used

- **Python:** data cleaning, preprocessing, automation

```
# dataset size
df.shape

df.info()

df.describe()

df.isnull().sum()

df.drop_duplicates(inplace=True)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   transaction_id         5000 non-null   object 
 1   customer_id            5000 non-null   object 
 2   transaction_date        5000 non-null   object 
 3   merchant_category      5000 non-null   object 
 4   channel                5000 non-null   object 
 5   device_id              5000 non-null   object 
 6   transaction_amount      5000 non-null   float64
 7   is_repeat_customer      5000 non-null   int64  
 8   days_since_last_txn     5000 non-null   int64  
 9   is_fraud                5000 non-null   int64  
10   age                    5000 non-null   int64  
11   gender                  5000 non-null   object 
12   region                  5000 non-null   object 
13   account_type            5000 non-null   object 
14   income_band             5000 non-null   object 
15   risk_score              5000 non-null   float64
16   risk_band               5000 non-null   object 
dtypes: float64(2), int64(4), object(11)
memory usage: 664.2+ KB
```

```
monthly = df.set_index('transaction_date').resample('M')['transaction_id'].count()

transaction_date
2025-02-28    609
2025-03-31    833
2025-04-30    843
2025-05-31    819
2025-06-30    753
Freq: M, Name: transaction_id, dtype: int64
```

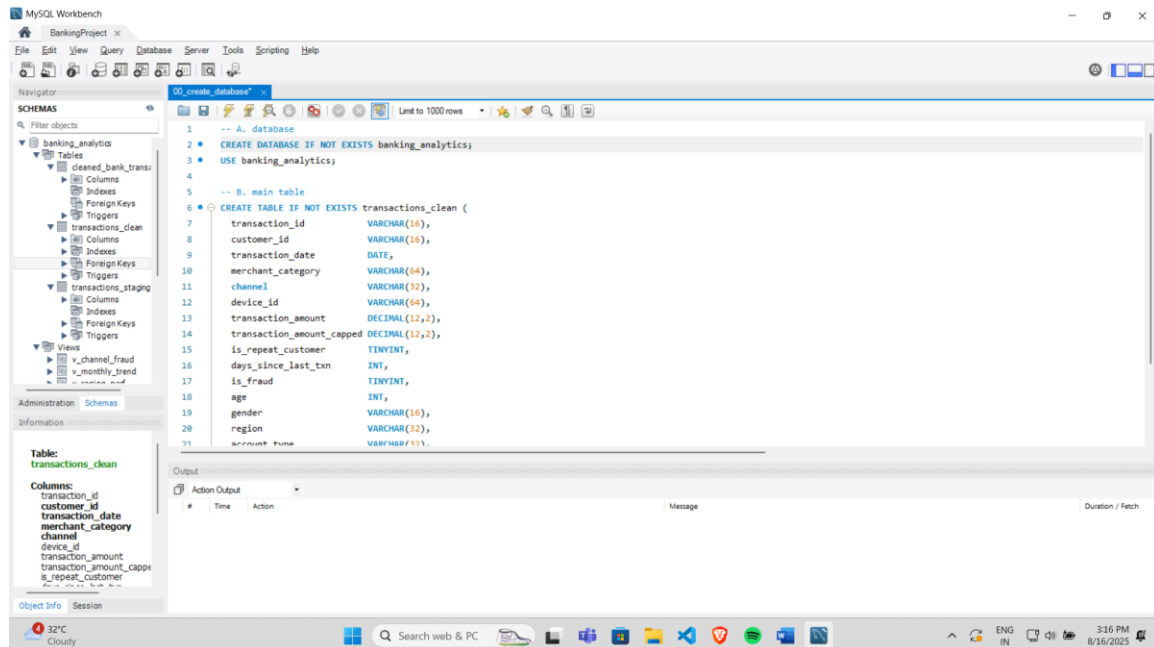
```
df.to_csv("../data/cleaned_bank_transactions.csv", index=False)
```

```
# region-month summary (super handy in Excel/Power BI)
summary = (
    df.assign(month=df['transaction_date'].dt.to_period('M').astype(str))
    .groupby(['region', 'month', 'risk_band'], as_index=False)
    .agg(
        transactions=('transaction_id', 'count'),
        total_amount=('transaction_amount_capped', 'sum')
    )
)

summary.to_csv("../data/summary_region_month.csv", index=False)
summary.head()
```

```
region  month  risk_band  transactions  total_amount
0   East  2025-02      High              2         2774.890
1   East  2025-02      Low             98        16445.650
2   East  2025-02    Medium             30        20552.000
3   East  2025-03      High              1         1387.445
4   East  2025-03      Low            154        27778.650
```

- **SQL:** Data storage, cleaning, and creation of analytical views.

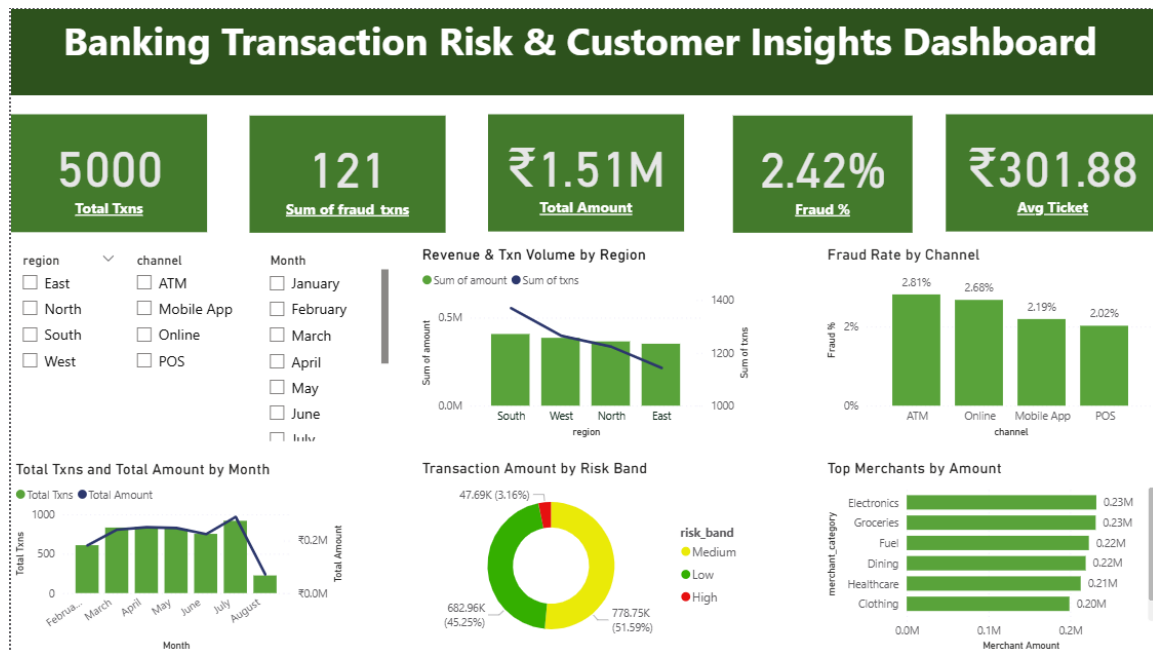


- **Excel:** Pivot tables and charts for preliminary decision logic analysis.

The screenshot shows the Microsoft Excel interface with a data table and a pivot table. The data table has columns: transaction_id, customer_id, merchant, channel, device_id, transaction_date, transaction_amount, transaction_amount_capped, is_repeat_customer, days_since_last_txn, is_fraud, age, gender, region, account_type, income_band, risk_score, risk_band, high_freq, and transaction_amount_capped. The pivot table is located in the bottom right corner and shows the following data:

Region	Gender	Age	Income Band	Risk Score	Risk Band	High Freq	Transaction Amount Capped
North	Male	56	>200K	517.03	Medium	1	430.86
North	Male	56	>200K	220.45	Low	1	183.71
North	Male	56	>200K	674.77	Medium	0	674.77
North	Male	56	>200K	1.35	Low	0	1.35
North	Male	56	>200K	438.78	Low	0	438.78
North	Male	56	>200K	19.82	Low	0	19.82
North	Male	56	>200K	367.79	Low	0	367.79
South	Male	69	>200K	556.7	Medium	1	463.92
South	Male	69	>200K	125.84	Low	0	125.84
South	Male	69	>200K	332.53	Low	0	332.53
West	Male	46	>200K	356.32	Low	1	296.93
West	Male	46	>200K	116.94	Low	0	116.94
West	Male	46	>200K	1387.45	Medium	0	1387.45
North	Female	32	<50K	221.65	Low	1	130.38
North	Female	32	<50K	279.5	Low	0	279.5
North	Female	32	<50K	667.44	Medium	1	556.2
North	Female	32	<50K	364.55	Low	0	364.55
North	Male	60	<50K	215.99	Low	1	179.99
West	Male	60	<50K	32.79	Low	0	32.79
West	Male	60	<50K	22.52	Low	0	22.52
West	Male	60	<50K	16.83	Low	0	16.83
South	Male	25	<50K	71.87	Low	1	59.89
South	Male	25	<50K	162.14	Low	0	162.14
South	Male	25	<50K	12.06	Low	0	12.06
South	Male	25	<50K	285.89	Low	0	285.89
South	Male	25	<50K	110.08	Low	0	110.08
South	Male	25	<50K	97.92	Low	0	97.92
North	Male	38	>200K	166.29	Low	1	138.90

- **Power BI:** Interactive dashboard for visualizing KPIs and generating insights.



Final Deliverable

The final output is an interactive Power BI dashboard that consolidates fraud monitoring, risk segmentation, regional performance, and top merchant analysis. This provides management with actionable intelligence to improve security, customer satisfaction, and profitability.